Exploring Recurrent Architectures for EMG-to-QWERTY Decoding

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Abstract

This report template is prepared for the ECE C147/C247 final project on the emg2qwerty dataset. Our project studies how recurrent sequence models decode typed characters from wrist electromyography (EMG) signals, with a primary focus on comparing vanilla RNN, LSTM, and GRU architectures under the provided single-user split for subject 89335547. The scaffold below is intentionally structured to match the required class write-up sections while leaving clear placeholders for final results, analysis, and reproducibility details. Dummy values are included in the main result tables so the report can be compiled and iterated on before all experiments are complete.

1 Introduction

The ECE C147/C247 final project emphasizes exploration beyond standard CNN pipelines, making the emg2qwerty dataset a natural testbed for sequential models. In this project, the core goal is to predict QWERTY keystrokes from surface electromyography signals recorded from both wrists. Because the input is continuous temporal data and the output is an unsegmented character sequence, the task is well suited to recurrent architectures trained with the connectionist temporal classification (CTC) loss [1].

Rather than re-introducing the general EMG decoding problem, this report should motivate the specific experimental question being studied. One suitable framing for this project is: *which recurrent architecture and training setup achieve the lowest character error rate on the provided single-user split?* Another angle is to ask how much performance changes with hidden size, depth, dropout, training duration, or available training data. The course rubric rewards not only performance, but also creativity and insight, so the final draft should explain why a design choice helps or hurts decoding quality rather than only listing metric values.

The remainder of this scaffold is organized around the required course sections: **Methods**, **Results**, and **Discussion**. It also includes placeholder tables for architecture comparisons, hyperparameter sweeps, official and future data splits, and compute reporting so the final paper can be updated quickly as experiments finish.

Table 1: Official single-user split used for the main experiments

Split	Sessions	User ID	Notes
Train	16	89335547	Official training sessions from the provided config
Validation	1	89335547	Used for model selection and early comparison
Test	1	89335547	Held out for the final architecture comparison

Table 2: Architecture configurations to compare

Model	Layers	Hidden size	Dropout	Decoder
RNN	3	256	0.10	CTC greedy
LSTM	3	256	0.10	CTC greedy
GRU	4	256	0.10	CTC greedy
TDS baseline	–	–	–	CTC greedy

2 Methods

2.1 Dataset and evaluation setup

This project uses the emg2qwerty dataset released by Meta Reality Labs [2]. For the baseline course setting, we use the provided single-subject split for user 89335547. The codebase already defines a train/validation/test split in `single_user.yaml`; the final report should state that these official sessions were used for the main comparison so results across architectures are directly comparable.

The main evaluation metric is character error rate (CER), which is the primary metric specified in the project instructions. It is also useful to report the deletion, insertion, and substitution components separately, since those often reveal different failure modes. If training logs already contain CTC loss, CER, IER, DER, and SER, the final paper should report those values consistently across validation and test splits.

2.2 Model families

Our current study compares several recurrent architectures that operate on windowed EMG features and decode character sequences with CTC. The minimum course expectation is to evaluate at least one recurrent model, but this report is prepared for a broader architecture study including vanilla RNNs, LSTMs, and GRUs. If time permits, the same structure can also include a baseline TDS model, a CNN+RNN hybrid, or a transformer-style model for additional comparison.

The final draft should briefly explain each model family, but only to the extent needed to justify the comparison. For example, one paragraph can note that vanilla RNNs provide a lightweight recurrent baseline, LSTMs explicitly model long-term memory through gating, and GRUs offer a simplified gating mechanism that may yield a better trade-off between optimization stability and compute cost.

2.3 Training protocol and hyperparameters

To make the report reproducible, list the optimizer, learning rate schedule, batch size, number of epochs, compute device, and any feature preprocessing choices. The scaffold below gives a compact place to summarize runs. Replace the dummy values with the exact settings used in the best and comparison models.

If you investigate preprocessing or data-modification strategies, add a short subsection here describing exactly what changed. Examples include channel selection, alternative normalization, reduced training fractions, or modified sampling rates. Keep the text focused on what was changed and why that change could plausibly affect CER.

Table 3: Sample hyperparameter log for recurrent model experiments

Run ID	Model	LR	Batch size	Epochs	Hidden size	Notes
A1	RNN	3e-4	32	40	256	Baseline recurrent run
A2	LSTM	3e-4	32	40	256	Same setup as A1
A3	GRU	3e-4	32	40	256	Best gating-based variant
A4	GRU	1e-4	32	40	384	Larger hidden state sweep
A5	GRU	3e-4	16	60	256	Longer training with smaller batch

Table 4: Sample architecture comparison with placeholder metrics

Model	Val CER ↓	Test CER ↓	Test IER	Test DER	Test SER
RNN	19.8	27.4	3.1	4.2	20.1
LSTM	18.3	25.6	2.9	3.7	19.0
GRU	17.6	24.9	2.7	3.5	18.8
TDS baseline	21.5	29.3	3.8	4.6	20.9

3 Results

This section should present the main quantitative findings and point the reader to the most important comparisons. A clean structure is to start with the primary architecture comparison on the official single-user split, then follow with one or two ablations such as hyperparameter sensitivity, reduced-data training, or alternative preprocessing.

Table 4 is currently populated with dummy values so the paper has a realistic layout during drafting. In the final version, replace these numbers with actual metrics from the training logs and state which model achieved the best validation CER and which model achieved the best held out test CER. If a model performs inconsistently across validation and test, discuss that explicitly rather than hiding it.

3.1 Potential split and data-efficiency studies

The course handout explicitly suggests studying how much data is needed, how many channels are needed, and how sampling rate affects performance. The table below is included so those additional experiments can be plugged into the final report without restructuring the document.

If these studies are included, the final draft should explain the split protocol precisely. For example, if you sample only a subset of train sessions, state whether the validation and test sessions remain unchanged. If the experiment changes channels or sampling rate, make clear whether the model architecture is otherwise held fixed.

3.2 Qualitative placeholders

If you later generate plots, this report is ready for:

- training and validation CER over epochs,
- CER as a function of hidden size or number of layers,
- CER versus fraction of training data,
- error breakdowns showing insertion, deletion, and substitution trends.

4 Discussion

The discussion should do more than restate the lowest CER. This is the section where you explain what you learned from the comparison. Questions worth answering include: Did gated recurrent models outperform the vanilla RNN? Was the improvement large enough to justify extra compute?

Table 5: Sample run summary with loss, CER, and compute placeholders

Run ID	Val loss	Val CER	Test loss	Test CER	Time (hr)	Device
A1	0.71	19.8	0.93	27.4	0.9	1×T4
A2	0.65	18.3	0.86	25.6	1.1	1×T4
A3	0.58	17.6	0.81	24.9	1.0	1×T4
A4	0.60	18.1	0.83	25.1	1.4	1×T4
A5	0.56	17.9	0.80	25.0	1.7	1×T4

Table 6: Suggested future split and ablation studies

Study type	Split or subset definition	Primary metric	Purpose
Reduced-data study	25%, 50%, 75%, and 100% of train sessions	CER	Measure data efficiency
Session ablation	First 4, 8, 12, and 16 train sessions	CER	Track how added sessions improve decoding
Channel ablation	Full channel set vs. reduced channel subsets	CER	Estimate minimum useful sensor count
Sampling-rate study	Downsampled EMG features at multiple rates	CER	Measure temporal resolution sensitivity
Cross-user extension	Train on one or more extra users, test per split	CER	Evaluate generalization beyond one user

Were some runs more stable during training? Did deeper or wider models actually help, or did they mostly overfit the single-user split?

For a strong final write-up, include at least one interpretation tied directly to the course rubric. For example, if the GRU achieves lower CER than the RNN, you might hypothesize that gating helps preserve relevant temporal context in the EMG sequence while avoiding some of the optimization difficulty of a plain recurrent model. If an LSTM does not improve over a GRU, you can discuss whether the added parameters and memory mechanisms were unnecessary for the observed time scale of the task.

4.1 Limitations

This project currently focuses on a single user because the baseline assignment explicitly allows full credit on that setting. However, that choice limits how strongly the results can generalize across users, sessions, and hardware conditions. The final discussion should also note any limits from compute budget, the number of random seeds used, and the extent of hyperparameter search. If error bars or repeated trials are not included, say so directly.

4.2 Reproducibility notes

The final report should include the exact training command, environment, and best checkpoint settings for the main comparison. One compact way to do this is to mention the training command in the discussion or appendix, for example using Hydra overrides such as `user=single_user model=gru_ctc trainer.max_epochs=40`. Add the exact command, seed, and device information once your final experiment recipe is settled.

Acknowledgments and Disclosure of Funding

This template follows the official NeurIPS 2024 style requested by the course handout and is adapted for the ECE C147/C247 final project on the emg2qwerty dataset. Replace this paragraph with any acknowledgments, compute-credit notes, or dataset access notes that you want in the final submission.

Figure 1: Placeholder for a training-curve or ablation plot. Replace this box with a real figure exported from your experiment notebook or analysis script.

Table 7: Compute and reproducibility checklist for final insertion

Item	Placeholder value	Final note to fill in
Framework	PyTorch Lightning	Version and environment file
Main device	1×GPU	Specify model, memory, and provider
Training time	~1 hr / 40 epochs	Replace with measured wall-clock time
Dataset split	Single user 89335547	Mention exact config path
Main command	`python -m emg2qwerty.train ...`	Insert exact command used

References

[1] Alex Graves, Santiago Fernandez, Faustino Gomez, and Jurgen Schmidhuber. Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376, 2006.

[2] Viswanath Sivakumar, Jeffrey Seely, Alan Du, Sean R. Bittner, Adam Berenzweig, Anuoluwapo Bolarinwa, Alexandre Gramfort, and Michael I. Mandel. *emg2qwerty: A large dataset with baselines for touch typing using surface electromyography*, 2024. arXiv:2410.20081.

[3] ECE C147/C247 staff. *ECE C147/C247, Winter 2026 project guidelines*, 2026.

A Appendix / supplemental material

Use the appendix for any material that supports the main claims but does not fit cleanly into the seven-page class report body. Good candidates include extra ablation plots, confusion analyses, exact hyperparameter grids, full reproduction commands, and additional tables for multi-user or cross-session experiments.

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